

M13-C3 — Probe catalogue (H1–H7 + H4b) motivated by closed H0

H0 closed; H1–H5 run/analysed;
H6–H7b run / analysed (not claims)

1 Status

Follow-up probes **motivated by closed baseline H0**, and **not** part of H0 itself. H0 is the completed six-cell AAMAS+ECAI investigation on fixture `m13_bucket3_binary_collider_and / n30_seed42`, documented in `learning_analysis.md / .tex` and summarised in `experiment.md`.

- H0: **closed** (see `learning_analysis.md`; not redefined here)
- H1: **run / analysed** — nested-prefix AAMAS ablation on `n25_seed42`. Record: `h1_support_ablation.md / .tex`. **Not a Bucket 3 claim.**
- H2: **run / analysed** — isolated- D AAMAS target c retains irrelevant d . Record: `h2_irrelevant_covariate.md / .tex`. **Not a Bucket 3 claim.**
- H3: **run / analysed** — BK-leading isolated a distracts ECAI target c from BD AND. Record: `h3_bk_leading_distractor.md / .tex`. **Not a Bucket 3 claim.**
- H4: **run / analysed** — repository `baseline_cautious` blocks brave root residual α -gadget on the H0 table. Record: `h4_cautious_vs_brave.md / .tex`. **Not a Bucket 3 claim.**
- H4b: **run / analysed** — on the H3 fixture, cautious changes only the α_3 close under shared leading- a gates (Layer B); Layer A shared with H3. Record: `h4b_cautious_split_under_a.md / .tex`. **Not H5. Not a Bucket 3 claim.**
- H5: **run / analysed** — experimental `greedy_cautious` matches AAMAS Greedy-brave on all 18 cells (Analysis A inert under Greedy). Record: `h5_greedy_cautious.md / .tex`. **Not a Bucket 3 claim. Not an AAMAS-paper config.**
- H6: **run / analysed** (`h6_folding_and_n_ablation.md`)
- H7a: **run / analysed** (`h7a_relto_vs_sechk.md`)
- H7b: **run / analysed** (`h7b_cautious_relto_vs_sechk.md`); not approved runs)
- **Not** approved Bucket 3 claims
- **Not** locked Bucket 1/2 claim content
- **Not** report-facing prose (`docs/report/findings/` is out of scope)
- Markdown twin: `future_probes.md`

Samuel decision (2026-08-01) — H1 / H2.

1. **H1 approved** — nested-prefix / missing-(0,0,0) AAMAS ablation on this fixture (e.g. $n = 25$, seed 42, targets a and b) against the approved $n = 30$ baseline.
2. **H2 design approved** — separate schema-2 fixture with isolated $D \sim \text{Bernoulli}(1/2)$ (variables a, b, c, d).
3. **H2c in scope** for the first H2 write-up.

Samuel decision (2026-08-01) — H3.

1. Graph/naming sketch **approved**.
2. Root parameters **approved**: $B \sim \text{Bernoulli}(4/5)$ ($P(B=1) = 4/5$, $P(B=0) = 1/5$); $D \sim \text{Bernoulli}(7/10)$ ($P(D=1) = 7/10$, $P(D=0) = 3/10$); $A \sim \text{Bernoulli}(1/2)$ as previously sketched.
3. Fixture id **locked**: `m13_bucket3_binary_bd_and_lead_a` (AND of parents `b,d`; BK-leading isolated `a`). Distinct from `m13_bucket3_binary_collider_and` and from the completed H2 fixture id `m13_bucket3_binary_collider_and_iso_d`.
4. **AAMAS contrast in scope** for the H3 write-up (secondary to primary ECAI inspection of target `c`; valuable as a related but distinct check from H2).

Samuel decision (2026-08-03) — H4 / H5.

1. Add a repository-baseline cautious target-wise configuration, named `baseline_cautious`, which explicitly pins the existing engine defaults and keeps `check_ic` for output generation.
2. Permit explicit `brave` and `cautious` modes in the target-wise runner while leaving the published ECAI and AAMAS configurations unchanged.
3. Prepare H4 on the closed H0 fixture and sample, with all targets generated automatically; roots (a)/(b) are primary and child (c) is a control.
4. Do **not** run H4 as part of the infrastructure change.
5. Defer any cautious Greedy ABA Learning design to **H5**. No `aamas_cautious` configuration is approved.

H1–H5 have been run and analysed (`h1_support_ablation.md`, `h2_irrelevant_covariate.md`, `h3_bk_leading_distractor.md`, `h4_cautious_vs_brave.md`, `h4b_cautious_split_under_a.md`, `h5_greedy_cautious.md`). Approvals do not create a Bucket 3 claim or a run matrix.

2 Motivation (verified facts from the AAMAS arm of closed H0)

On the approved frozen sample `n30_seed42`, configuration `aamas2025` (`greedy / folding_selection(mgr) / folding_space(bk) / brave / check_ic`):

Target	Outcome	Delta (if any)
<code>c</code>	<code>solved</code>	<code>c(A) :- a_val_1(A), b_val_1(A).</code> (assumption-free)
<code>a</code>	<code>completed_no_solution</code>	empty
<code>b</code>	<code>completed_no_solution</code>	empty

Sources:

- collection summary: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/summary.md`
- target `c` delta: `.../cells/target-c/output/delta.aba`
- target `a/b` reports and `prolog.stdout` under `.../cells/target-{a,b}/`

For target `c`, the learned delta string coincides with the evaluator-only positive-state reference in `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/mechanism_reference.json`. Record that as **syntactic coincidence**, not as a causal-recovery verdict.

Procedural reading of the AAMAS root-target path (verified in `.../cells/target-a/output/prolog.stdout`; target `b` is symmetric):

1. Greedy folding builds **maximal co-occurrence bodies** from BK for each E^+ row (not a minimal-parent search).
2. The both-1 co-occurrence body $a(A) :- b_val_1(A), c_val_1(A)$ is accepted under brave entailment.
3. The both-0 co-occurrence body $a(A) :- b_val_0(A), c_val_0(A)$ over-generalises; assumption/contrary repair is attempted.
4. Contrary rote learning cites rows **26** and **30** — the only sample atoms $(0,0,0)$ in `samples/n30_seed42.csv` — and contrary folding then fails, yielding `completed_no_solution`.

These facts from closed H0 motivate probe families H1 and H2. They do not by themselves approve a Bucket 3 claim.

2.1 Motivation addendum (ECAI arm of closed H0 — for H3)

On the same frozen sample `n30_seed42`, configuration `ecai2024 (folding_mode(nd), folding_selection(any), brave / check_ic)`, target c **solved** with delta (verified):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A), c_alpha_1(A)) :- assumption(alpha_1(A)).
```

In `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout`, the first `nd` fold on an early E^+ row replaces the sample id by `a_val_1(A)` (the first BK predictor under declared order `a, b` for that cell), then assumption repair retains a in the ordinary target body while the second parent enters via a contrary. Collection summary: `.../ecai2024/n30_seed42/summary.md`.

This motivates **H3**: put an *independent* distractor first in BK by naming, under ECAI, on a new fixture. H3 is now **run / analysed** (`h3_bk_leading_distractor.md`); it is still **not** a Bucket 3 claim.

3 Ordering

1. **H1** — nested-prefix / missing- $(0,0,0)$ ablation on the existing fixture and AAMAS config (seed 42). **Run / analysed** (`h1_support_ablation.md`).
2. **H2** — separate schema-2 fixture with isolated covariate D on the $A \rightarrow C \leftarrow B$ AND collider. **Run / analysed** (`h2_irrelevant_covariate.md`). Lead: AAMAS c retains `d_val_*`.
3. **H3** — separate schema-2 fixture `m13_bucket3_binary_bd_and_lead_a` with BK-leading independent A and AND collider $B \rightarrow C \leftarrow D$. **Run / analysed** (`h3_bk_leading_distractor.md`). Lead: ECAI first-folds to a and retains a in the theory for c .
4. **H4** — same fixture/`n30_seed42`; locked ECAI brave vs repository `baseline_cautious`; roots a/b primary; c control. **Run / analysed** (`h4_cautious_vs_brave.md`). Lead: cautious KO on brave residual α -reuse; roots `completed_no_solution`; control c still solves.
5. **H5** — experimental `greedy_cautious` counterparts of all completed AAMAS collections. **Run / analysed** (`h5_greedy_cautious.md`). Lead: brave $\rightarrow$ cautious under Greedy is inert on these 18 cells.
6. **H6–H7b (run / analysed)** — H6 and H7 were prompted by H4 and are recorded separately from H4.

Do not expand any family into a broader run matrix beyond these bounded probes without a further Samuel decision.

3.1 Relation among probe families (keep distinct)

Probe	Focus
H1	AAMAS roots; missing negative witness / both-0 over-generalisation on the current three-variable fixture
H2	AAMAS target <i>c</i> ; irrelevant covariate enters greedy maximal co-occurrence bodies (minimality vs solvability); fixture id <code>m13_bucket3_binary_collider_and_iso_d</code>
H3	ECAI target <i>c</i> (primary); BK order puts independent distractor first ; nd first-fold / repair path; AAMAS contrast in scope ; id <code>m13_bucket3_binary_bd_and_lead_a</code>
H4	Same fixture/sample ; locked ECAI brave vs repository <code>baseline_cautious</code> ; roots <i>a/b</i> primary; <i>c</i> control; run / analysed
H5	Experimental <code>greedy_cautious</code> vs AAMAS brave on 18 cells; Analysis A inert; run / analysed ; not AAMAS-paper config
H6	Run / analysed; length drivers under cautious+nd (H6a/H6b);
H7a	Run / analysed; brave nd relto vs sechk;
H7b	Run / analysed; cautious nd <code>asm_intro(sechk)</code> vs relto

H3 is not a duplicate of H2: different DAG, different distractor placement/name, different primary config (ECAI nd vs AAMAS greedy), and distinct fixture ids. The in-scope AAMAS arm on H3 still differs from H2 because the irrelevant variable is BK-leading *a* and the mechanism parents are *b, d*.

H4 is not a duplicate of H1 or H3: it holds the table fixed and changes the learning mode while retaining the engine’s baseline nd/any/all/relto options. H5 was kept separate because a cautious Greedy learner needed a paper- and algorithm-level specification before its experimental configuration could be interpreted.

4 Probe family H1 — support sensitivity on AAMAS root targets

Status: **run** / **analysed** / not a claim. Evidence record: `h1_support_ablation.md` / `h1_support_ablation.tex`.

Evaluative stance (locked for H1 reading). Roots have `no_observed_parent_deterministic_rule`. H0 AAMAS root `completed_no_solution` is the **desired** / **correct** outcome. H1 AAMAS root solved with co-occurrence Horn rules is an **incorrect** / **spurious** finite-sample solution.

4.1 H1a — spurious root rule under missing negative witness

On this fixture, AAMAS, target *a* (symmetrically *b*), seed 42: the nested prefix $n=25$ **omits all** $(0,0,0)$ **rows**. Learning **solved** with a delta that includes the both-0 co-occurrence rule

`a(A) :- b_val_0(A), c_val_0(A).`

(respectively `b(A) :- a_val_0(A), c_val_0(A).`), which is **false** as a population implication and is blocked on the closed H0 $n=30$ table by E^- rows 26 and 30.

Nested-prefix fact (seed 42): rows 1–25 contain no $(0,0,0)$ atom; rows 26 and 30 are the only $(0,0,0)$ observations in $n=30$. Verified: `n25_seed42.csv` is a row-for-row prefix of `n30_seed42.csv`.

4.2 H1b — negative witness in the unsafe cell

If the sample contains at least one opposite-label row in that both-0 predictor cell, the both-0 rule fails brave entailment under the observed AAMAS path. On closed H0 $n=30$ this manifested as `completed_no_solution`, not as emission of only the safe both-1 rule. That H0 outcome is the **correct** reading for roots under the H1 stance.

4.3 Explicit non-claims for H1

- Full support does **not** imply AAMAS will emit a “correct” deterministic root mechanism rule: roots have `no_observed_parent_deterministic_rule` in the evaluator reference.
- “One of each population atom” is a sufficient practical condition on this approved sample, but the operational point is a **label conflict inside the greedy candidate body cell**, not a generic demand for full joint support.
- H1 does not claim that omitting (0,0,0) is the only way AAMAS can emit a spurious root rule, nor that every seed behaves identically.
- H1 does **not** claim that root `solved` on $n=25$ is progress or mechanism recovery.

4.4 Outcome (H1)

Run / analysed. Nested-prefix ablation executed on `m13_bucket3_binary_collider_and / AAMAS / seed 42 / n=25`. Roots a/b **incorrectly** solved with both-1 and both-0 co-occurrence rules. Child c remained AND-solved as a control. Full narrative: `h1_support_ablation.md`.

5 Probe family H2 — irrelevant isolated covariate on AAMAS target c

Status: **run / analysed** / not a claim. Evidence record: `h2_irrelevant_covariate.md / h2_irrelevant_covariate.tex`.

Lead finding. On fixture `m13_bucket3_binary_collider_and_iso_d / n30_seed42`, AAMAS solves target c with

```
c(A) :- a_val_1(A), b_val_1(A), d_val_1(A).  
c(A) :- a_val_1(A), b_val_1(A), d_val_0(A).
```

so **irrelevant D is included**. The evaluator-only reference remains the D -free AND `c(A) :- a_val_1(A), b_val_1(A) ..` Failure mode: distractor retention / minimality, not unlearnability.

5.1 H2a — irrelevant covariate enters greedy bodies

Observed as predicted: when both values of D appear among E^+ for c (9 each on $n=30$), greedy catalogue bodies **include** `d_val_*`.

5.2 H2b — learnability vs minimality

c **still solved** (two D -conditioned rules). Confirms that the interesting failure is retention of an irrelevant predictor, not `completed_no_solution`.

5.3 H2c — sample-edge (also run; not the centre)

Nested $n=2$ prefix where $C=1$ realises only $d=1$: single rule $c :- a_val_1, b_val_1, d_val_1$. Extreme edge. Documented briefly in `h2_irrelevant_covariate.md`. Not the scientific centre of H2.

5.4 Outcome (H2)

Run / analysed. Fixture authored and certified; AAMAS `n30_seed42` primary target c shows distractor inclusion; H2c $n=2$ also run. Full record: `h2_irrelevant_covariate.md`.

6 Probe family H3 — ECAI first-fold distraction by BK-leading independent variable

Status: **run / analysed** / not a claim. Evidence record: `h3_bk_leading_distractor.md` / `h3_bk_leading_distractor.tex`.

Lead finding. On fixture `m13_bucket3_binary_bd_and_lead_a` / `n30_seed42`, ECAI first-folds to `a_val_*` and retains a in both target rules for c . The emitted delta does **not** coincide with the evaluator-only $BD \text{ AND } c(A) :- b_val_1(A), d_val_1(A)$. Failure mode: BK-leading distractor inclusion / first-fold distraction, not unlearnability.

6.1 Fixture (as run)

Field	Value
Fixture id	<code>m13_bucket3_binary_bd_and_lead_a</code>
Internal / display	<code>a,b,c,d</code> / A, B, C, D
DAG	$B \rightarrow C \leftarrow D$; no edges involving A
A	Bernoulli(1/2); independent of $\{B, C, D\}$
B	Bernoulli(4/5) — $P(B=0) = 1/5, P(B=1) = 4/5$
D	Bernoulli(7/10) — $P(D=0) = 3/10, P(D=1) = 7/10$
C	deterministic $C = B \wedge D$
Evaluator reference for c	<code>c(A) :- b_val_1(A), d_val_1(A)</code> . (a must not appear)
BK order for target c	predictors a, then b, then d (irrelevant first)
Frozen sample	<code>n30_seed42</code>
Primary learning arm	ECAI target c
Secondary arm	AAMAS on the same frozen sample (in scope)

6.2 Scientific question (answered)

When learning target c under **ECAI** on this frozen sample, does BK-leading independent a **distract** the first fold and/or the final ABA shape away from the mechanism-aligned B, D conjunction?

Answer (bounded): yes — first fold selects `a_val_1`; both target rules retain a ; metrics body vars $\{a\}$. Secondary: AAMAS also retains `a_val_*` in both Horn bodies (vs H2's trailing- d inclusion).

6.3 Secondary finding (in scope; not the lead)

Brave residual / underdetermining delta on the nested $a=1$ / $\alpha_1-\alpha_3$ side. Cross-link H0 ECAI target- a rows 9 vs 26. Reinforces: **solved** + **audit SAT** $\neq$ **mechanism-aligned**.

6.4 Explicit non-claims for H3

- Inclusion of a is **not** causal recovery of an $A \rightarrow C$ edge.
- Runner **solved** is **not** automatically correct / mechanism-aligned.
- H3 does not reuse H2's graph; distractor here is A , parents B, D .
- Completing H3 does **not** approve a Bucket 3 claim.

6.5 Outcome (H3)

Run / analysed. Fixture authored and certified; ECAI n30_seed42 primary target c shows BK-leading a distraction; AAMAS contrast also retains a . Full record: h3_bk_leading_distractor.md.

7 Probe family H4 — repository-baseline cautious vs published ECAI brave

Status: **run / analysed** / not a claim. Evidence record: h4_cautious_vs_brave.md / h4_cautious_vs_brave.tex.

Lead finding. On the closed H0 table m13_bucket3_binary_collider_and / n30_seed42, switching from locked **ECAI brave** to repository **baseline_cautious blocks the brave residual-assumption gadget** that had allowed roots a/b to **solved**. Paths coincide through early $\alpha_1-\alpha_3$ setup; at the closing step **c_alpha_3 :- alpha_2**, ... brave says **OK** and finishes, cautious says **KO**, then exhausts the folding budget and returns **completed_no_solution** (no delta). Control c still solves with a byte-identical delta to locked ECAI c .

7.1 Fixture / arms (as run)

Field	Value
Fixture / sample	m13_bucket3_binary_collider_and / n30_seed42
Sample hash	sha256:1edae465...a8ca
Brave arm	locked ecai2024
Cautious arm	baseline_cautious (configs/baseline_cautious_config.pl)
Primary	roots a, b
Control	c

7.2 Scientific question (answered)

Does repository-baseline cautious learning change the ECAI-brave root outcomes that relied on a residual per-row assumption gadget (rows 9 vs 26), and what happens to control c ?

Answer (bounded): yes for roots — cautious KO at the brave closing α -reuse; **completed_no_solution**. Control c still **solved** with identical delta.

7.3 Explicit non-claims for H4

- Not claimed that cautious mode is “better” for causal recovery.
- Not claimed that cautious root no-solution recovers a correct root mechanism (none exists).
- Not a ranking of brave vs cautious as general ABALearn policy.
- Control *c* string match / **solved** is not causal recovery.
- Under cautious learning, SAT of `bk.sol_chk.asp` is an existential final-artefact integrity witness, not a cautious-consequence proof.

7.4 Outcome (H4)

Run / analysed. Collection at `.../baseline_cautious/n30_seed42/`: *a/b* = `completed_no_solution`; *c* = `solved` (identical delta to ECAI). Full record: `h4_cautious_vs_brave.md`.

7.5 Prompted follow-ups (signpost only; not H4 evidence)

H5 is now **run / analysed** separately (`h5_greedy_cautious.md`). H6 is now **run / analysed** (`h6_folding_and_n_ablation.md`). H7a is now **run / analysed** (`h7a_relto_vs_sechk.md`). H7b is **run / analysed** separately (`h7b_cautious_relto_vs_sechk.md`).

8 Probe family H4b — cautious α_3 close under H3 leading-*a* gates

Status: **run / analysed** / not a claim. **Not H5.** Evidence record: `h4b_cautious_split_under_a.md` / `h4b_cautious_split_under_a.tex`.

Lead finding (Layer B). On H3 fixture `m13_bucket3_binary_bd_and_lead_a / n30_seed42` target *c*, repository `baseline_cautious` keeps H3’s leading-*a* gates and the persistent α_1 -nest vs α_2 -flat contrary split; it **only** changes the α_3 close — rejecting brave residual `c_alpha_3 :- alpha_1, a_val_1` and accepting `c_alpha_3 :- d_val_1`. Both arms **solved**.

Layer A (shared with H3; not a cautious effect): *a*-gating of target rules; nested `c_alpha_1` vs flat `c_alpha_2`; procedural reason is search order / **relto** / cover-and-repair (see the H4b record).

8.1 Outcome (H4b)

Run / analysed. Collection: `.../m13_bucket3_binary_bd_and_lead_a/baseline_cautious/n30_seed42/`; target *c* **solved** (~2.9s; audit SAT). Incidental: *a/b/d* timed out (outside focus). Full record: `h4b_cautious_split_under_a.md`.

9 Probe family H5 — experimental Greedy + cautious (greedy_cautious)

Status: **run / analysed** / not a claim. Evidence record: `h5_greedy_cautious.md` / `h5_greedy_cautious.tex`.

Lead finding (Analysis A). On all 18 paired cells, experimental `greedy_cautious` matches locked `aamas2025` on every outcome and every solved delta (byte-identical). Traces are isomorphic aside from entailment-logging surface differences. Flipping brave→cautious inside the Greedy

bundle does not change accepted or rejected theories on these tables. Mean runtime $\approx 1.84\times$ (still sub-second).

Secondary (Analysis B). Large contrasts appear only vs repository `baseline_cautious` (nd bundle) on H0/H3 $n=30$ — **search-strategy**, not the brave $\rightarrow$ cautious flip. Not a single-knob `folding_mode` ablation.

Config: `configs/greedy_cautious_config.pl` (experimental; not published AAMAS; do not call it `aamas_cautious`).

9.1 Outcome (H5)

Run / analysed. Full narrative: `h5_greedy_cautious.md`.

10 Probe family H6 — folding-steps and nested- n search-cost ablation

Status: **run** / **analysed** / not a claim. Evidence: `h6_folding_and_n_ablation.md` / `.tex`.

H6a: fixed `n30_seed42`; `folding_steps(M)  $\in \{1, 2, 5, 10\}$` . Root trace length $\approx$ linear in M via nd failed-band replays ($\text{NEW} = 9 \times M$); control c solves inside `tokens(1)`.

H6b: fixed $M=2$; nested $n \in \{30, 60, 90\}$. Topology fixed ($\text{NEW}=18$, $\text{KO}=32$); length $\approx$ linear in n via per-band bookkeeping.

Cross: H4's $\sim 5\text{k}$ -line root no-sols are primarily cumulative token-budget replays (H6a) with secondary n -scaling at fixed $M=2$ (H6b). Not a claim; not an $n \times \text{steps}$ interaction result.

10.1 Outcome (H6)

Run / analysed. Full narrative: `h6_folding_and_n_ablation.md`.

11 Probe family H7a — brave nd relto vs sechk

Status: **run** / **analysed** / not a claim. H7b analysed separately (`h7b_cautious_relto_vs_sechk.md`). Evidence: `h7a_relto_vs_sechk.md` / `.tex`.

Lead (contrast). On all seven H0+H3 pairs, both arms **solved** but every delta differs. Relto tries relative reuse first and often KOs into later fold literals; sechk always mints a provisional α then `gen5`-checks, often never taking those later-covariate folds. Under sechk: first-BK latch + α -chain; under relto: richer contrary covariates (esp. for c).

11.1 Outcome (H7a)

Run / analysed. Full narrative: `h7a_relto_vs_sechk.md`.

12 Probe family H7b — cautious nd relto vs sechk

Status: **run** / **analysed** / not a claim. Evidence: `h7b_cautious_relto_vs_sechk.md` / `.tex`.

Lead (Q1–Q2). On child c , both arms **solved** but deltas differ under cautious nd. Same **gen.pl** fork as H7a; cautious sechk retains b/d on c (unlike brave H7a). Q4–Q5 empty-stdout timeouts are non-comparable.

12.1 Outcome (H7b)

Run / analysed. Full narrative: `h7b-cautious_relto_vs_sechk.md`.

13 Boundary distinctions to preserve

Keep separate throughout any future probe write-up:

Object	Role
AAMAS strategy / greedy maximal co-occurrence search	procedural learner behaviour (H1/H2)
ECAI nd first-fold / BK serialisation order	procedural learner behaviour (H3)
Brave vs cautious entailment / learning mode	procedural / semantic learner behaviour (H4)
Cautious Greedy specification / experimental greedy_cautious	procedural / semantic probe (H5); not an AAMAS-paper config
Finite-sample support and label conflicts in candidate body cells	sample information
Mechanism correspondence for c (AND)	evaluator-only interpretation of a non-root
Absence of deterministic root mechanisms	evaluator reference status
Syntactic delta = reference string	coincidence, not automatic causal verdict
Runner solved under brave	brave coverage of $E^\pm$, not mechanism recovery

Do not treat predictive rules for roots as mechanism recovery.

14 Cross-links

- Investigation record: `experiment.md`
- Mathematical fixture dossier: `fixture_dossier.tex`
- Bucket 3 planning hub: `../M1.3-bucket3-claims.md` (still **no claim**)
- Research decision log: `docs/research/decisions.md` (2026-08-01 H1–H3 entries; 2026-08-03 H4–H6 entries)
- Fixture pre-run bundle: `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/`
- AAMAS baseline: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/`
- ECAI baseline (H3/H4 motivation): `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/ecai2024/n30_seed42/`
- Repository cautious baseline: `configs/baseline_cautious_config.pl`
- H4 config: `causal/configs/targetwise/m13_bucket3_binary_collider_and/baseline_cautious/n30_seed42.yaml`

- H4 collection: `causal/outputs/aba_learning/targetwise/ m13_bucket3_binary_collide_r_and/baseline_cautious/n30_seed42/`
- Target-wise explicit brave/cautious validator: `causal/targetwise/config.py`
- Entailment implementation: `asp_engine.pl` (entails/5)

15 Decisions recorded

Item	Decision	Date
H1 nested-prefix / missing-(0,0,0) AAMAS ablation	Approved then run / analysed	2026-08-01 / 2026-08-02
H2 four-variable isolated- D fixture + AAMAS c	Approved then run / analysed	2026-08-01 / 2026-08-02
H2c in first H2 write-up	In scope (run; brief note only)	2026-08-01 / 2026-08-02
H3 graph/naming; B 80/20, D 70/30; id m13_bucket3_binary_bd_and_lead_a; AAMAS contrast	Design approved then run / analysed	2026-08-01 / 2026-08-02
H4 repository-baseline cautious vs published ECAI brave	Infrastructure ready then run / analysed	2026-08-03
H4b cautious α_3 close on H3 leading- a target c	Run / analysed (not H5)	2026-08-03
H5 experimental greedy_cautious (18 cells)	Run / analysed	2026-08-03
H6 folding/ n ablation (H4-prompted)	Approved then run / analysed	2026-08-03
H7a brave nd sechk vs relto	Approved then run / analysed	2026-08-04
H7b cautious nd sechk vs relto	Approved then run / analysed	2026-08-04

16 Current status and remaining work

- **H1: complete** (h1_support_ablation.md). Not a claim.
- **H2: complete** (h2_irrelevant_covariate.md). Lead: AAMAS c retains irrelevant d . Not a claim.
- **H3: complete** (h3_bk_leading_distractor.md). Lead: ECAI c first-folds to / retains BK-leading a (not BD AND). Not a claim.
- **H4: complete** (h4_cautious_vs_brave.md). Lead: cautious blocks brave root residual α -gadget; roots **completed_no_solution**; control c still solves. Not a claim.
- **H4b: complete** (h4b_cautious_split_under_a.md). Lead (Layer B): only **c_alpha_3** close changes under shared Layer A. Not a claim; not H5.
- **H5: complete** (h5_greedy_cautious.md). Lead: Greedy brave→cautious inert on 18 cells; Analysis B contrasts are search-bundle. Not a claim; not AAMAS-paper config.
- **H6: complete** (h6_folding_and_n_ablation.md). Lead: root cost $\approx$ linear in **folding_steps**(M) and nested n at $M=2$. Not a claim.
- **H7a: complete** (h7a_relto_vs_sechk.md). Lead: 7/7 solved with differing deltas; **sechk** first-BK latch vs richer **relto** contraries. Not a claim.
- **H7b: complete** (h7b_cautious_relto_vs_sechk.md). Lead: Q1–Q2 c deltas differ; Q4–Q5 non-comparable. Not a claim.
- Still **no** Bucket 3 claim and **no** expanded run matrix without a further Samuel decision.